

High-Performance Acceleration of 2D and 3D CNNs on FPGAs using Static Block Floating-Point

Hongxiang Fan, Shuanglong Liu, Zhiqiang Que, Xinyu Niu, Wayne Luk, *Fellow, IEEE*,

Abstract—Over the past few years, two-dimensional (2D) convolutional neural networks (CNNs) have demonstrated their great success in a wide range of 2D computer vision applications such as image classification and object detection. At the same time, three-dimensional (3D) CNNs, as a variant of 2D CNNs, have shown their excellent ability to analyse 3D data such as video and geometric data. However, the heavy algorithmic complexity of 2D and 3D CNNs imposes a substantial overhead over the speed of these networks, which limits their deployment in real-life applications. Although various domain-specific accelerators have been proposed to address this challenge, most of them only focus on accelerating 2D CNNs, without considering their computational efficiency on 3D CNNs. In this paper, we propose a unified hardware architecture to accelerate both 2D and 3D CNNs with high hardware efficiency. Our experiments demonstrate that the proposed accelerator can achieve up to 92.4% and 85.2% multiply-accumulate (MAC) efficiency on 2D and 3D CNNs respectively. To improve the hardware performance, we propose a hardware-friendly quantization approach called static block floating-point (BFP) which eliminates the frequent representation conversions required in traditional dynamic BFP arithmetic. Comparing with the integer linear quantization using zero-point, the static BFP quantization can decrease the logic resource consumption of the convolutional kernel design by nearly 50% on an FPGA. Without time-consuming retraining, the proposed static BFP quantization is able to quantize the precision to 8-bit mantissa with negligible accuracy loss. As different CNNs on our reconfigurable system require different hardware and software parameters to achieve optimal hardware performance and accuracy, we also propose an automatic tool for parameter optimization. Based on our hardware design and optimization, we demonstrate that the proposed accelerator can achieve 1.5 ~ 5.6 times higher energy efficiency than GPU implementation. Comparing with the state-of-the-art FPGA-based accelerators, our design achieves higher generality and up to 1.4 ~ 2.2 times higher resource efficiency on both 2D and 3D CNNs.

Index Terms—Three-dimensional convolutional neural network (3D CNN), field-programmable gate array (FPGA), static block floating point (BFP)

This work was supported in part by the United Kingdom EPSRC under Grant EP/L016796/1, Grant EP/N031768/1, Grant EP/P010040/1, Grant EP/V028251/1 and Grant EP/S030069/1, the National Natural Science Foundation of China (No. 62001165), Hunan Provincial Natural Science Foundation of China (No. 2021JJ40357), Changsha Municipal Natural Science Foundation (No. kq2014079), Shenzhen Science and Technology Program (Project No. KQTD20200820113009020) and in part by the funds from Corerain, Maxeler, Intel, Xilinx and SGIIT. Corresponding author: Shuanglong Liu (liu.shuanglong@hunnu.edu.cn).

H. Fan, Z. Que and W. Luk are with the Department of Computing, Imperial College London, London, SW7 2AZ, UK.

S. Liu is with the School of Physics and Electronics, Hunan Normal University, Changsha 410081, China.

X. Niu is with Corerain Technologies Ltd., Shenzhen, China.

I. INTRODUCTION

In recent years, deep neural networks (DNNs), especially convolutional neural networks (CNNs), have demonstrated their great potential in various computer vision (CV) applications. In particular, 2-Dimension (2D) CNNs, which perform 2D convolution in spatial domain to extract 2D features, have achieved the state-of-the-art accuracy in a wide range of CV tasks including image classification [1] and object detection [2]. Besides 2D CNNs, 3-Dimension (3D) CNNs [3], due to their ability of incorporating the third-dimensional information based on 3D convolution, have been also adopted in various 3D CV scenarios such as human action recognition [4] and 3D medical imaging segmentation [5].

Nevertheless, the memory and computational complexity of 2D and 3D convolutions put a heavy burden on their hardware performance on general-purpose processors, which restrains their application in real-life scenarios [6]. For instance, a classical 3D CNN designed for human action recognition called *C3D* requires nearly 78 GOPs, and thus achieves only 951ms per inference for a 16-frame video on an Intel i5 CPU, which cannot meet the requirement of real-time processing [7]. Therefore, there is a great demand of domain-specific accelerators for both 2D and 3D CNNs. Different hardware platforms including GPUs, ASICs and FPGAs have been used to accelerate both 2D and 3D CNNs. Among all these hardware platforms, FPGAs are gaining popularity because of their better flexibility than ASICs and higher energy efficiency than GPUs [8], [9]. In spite of these advantages, there are several challenges when accelerating both 2D and 3D CNNs on FPGA:

- To improve the hardware performance, most of FPGA-based accelerators tend to utilize low-precision weights or activations [8]. However, previous work either shows significant accuracy loss, such as fixed-point [10] and logarithm arithmetic [11], or requires a large amount of hardware resource on implementing quantization module, such as linear quantization [12] and dynamic BFP quantization [13].
- Convolution operations, especially 3D convolution, are highly memory and computation-intensive, making it hard to achieve high performance on FPGAs with limited memory and computational resources [7].
- Unifying 2D and 3D CNNs onto one hardware architecture is challenging since they have different data locality, and improper design may cause high degree of data replication [14].

To address these challenges, this paper proposes a high-performance reconfigurable architecture for accelerating 2D